Group Project

20 Points Possible

6/9/2026

Attempt 1

In Progress
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4/30/2026 to 6/9/2026

Details

Overview

Quantitatively understanding the statistical significance of scientific results requires us to perform a mathematical assessment of evidence from experiments. This small group project will teach us

1) How to model a scientific experiment that includes an underlying relationship with both intrinsic randomness and experimental uncertainty.

2) Determine the significance of a measurement that deviates from the intrinsic randomness expected from an experiment.

3) Program Python functions to compute mathematical models and simulate experimental noise.

4) Plot models and data on the same graph using a Jupyter notebook, and provide a graphical representation of the statistical significance of an experimental event.

5) Develop code collaboratively using a standard development platform (GitHub).

Instructions

For our small group project, we will be applying data reduction to an exoplanet transit and determining the exoplanet's radius relative to its host star, as well as its period around the star. Please sign up for a Group based on your discussion section **here (https://docs.google.com/spreadsheets/d/1aVLV1ie3DVbvkY_Dz35gW85lNbFFmlS0g1UuN4ivaUk/edit?usp=sharing)**. I will finish assigning people to groups **May 8th**, so if you do not assign yourself to one until then, I will.

1) Download the **Planet_Lightcurve.fits** file on Canvas. This lists the relative flux (or brightness) of the star observed over a time range of a couple of days. Orbiting around this star is an exoplanet, and

in this dataset it passes in front of its host star, blocking some of the star's light: this is called a transit. However, the dataset is noisy, and has both linear and oscillatory artifacts, which must be removed.

2) Using either the curve_fit function from the scipy library or your own fitting algorithm, come up with a way to fit for and divide out both the linear and the oscillatory artifacts in the data, assuming the root mean squared experimental error on the flux measurements is 0.003. Document these steps in the Jupyter notebook, making sure to report the uncertainty in your best fit parameters, and graph both the model and the data on the same plot. Also plot what the data looks like after the artifacts have been removed. Label each of your plots to provide clarity, and have the notebook save the figures as PDFs.

3) Next, using your cleaned-up dataset, have your code come up with an estimate for both the depth of the transit in units of relative flux and for its length in days. Using the transit depth D, you can estimate the radius of the exoplanet using the following equation (assume the radius of the star is one solar radii):

$$D = \left(\frac{R_{planet}}{R_{star}}\right)^2$$

4) Using error propagation, find out what the root mean squared uncertainty in your measurement of D and the transit time is. Don't forget to take into account both the uncertainty in the data **and** the uncertainty from your fits.

5) Stars can sometime experience a solar flare, which is a sudden increase in brightness as it ejects material from its surface. Some solar flares can cause the star to become 10% brighter in a short period of time. How large of a deviation is a solar flare of this magnitude, in terms of standard deviations of the distribution of stellar flux from your cleaned data? Add a single 10% outlier somewhere along your lightcurve that the transit isn't taking place. Replot the data as part of your notebook, report the N-sigma statistical significance of this event. Using Chauvenet's criterion, would this outlier be significant enough to be worthy of attention?

Procedure

This group assignment (max 5 students per group) will require you to develop and turn in one Jupyter notebooks. The Jupyter notebook should be hosted as a project on GitHub where every member of the group has "collaborator" permissions to the repository. Each member of the group should contribute Python code to the project, demonstrating their contributions using the branch / pull request mechanism in GitHub. This procedure will automatically document the contribution of each student to the group assignment. You should divide members to work on each of the 4 main tasks in this project (tasks 2-5). Much of the architecture required can be programmed even before the data necessary is ready, so all students should be working from the beginning, even if they are focusing on the last task.

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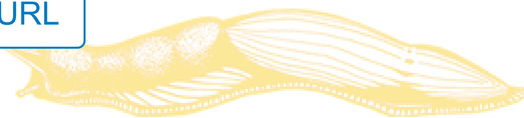

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